CBSE Class 10 Mathematics

Time Allowed: 3 Hours **Maximum Marks: 10**

General Instructions:

1. This question paper contains 10 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$, where x and y are prime numbers, then the $HCF(a, b)$ is:

(A) (A) x^3y^3
(B) (B) xy^2
(C) (C) x^2y^2
(D) (D) x^3y^3

2. The decimal expansion of the rational number $\frac{43}{2^4 \times 5^3}$ will terminate after how many places of decimals?

(A) (A) 2
(B) (B) 3
(C) (C) 5
(D) (D) 4

3. If the HCF of 65 and 117 is expressible in the form $65m - 117$, then the value of m is:

(A) (A) 1
(B) (B) 3
(C) (C) 2
(D) (D) 4

4. The largest number which divides 70 and 125 leaving remainders 5 and 8 respectively is:

(A) (A) 13
(B) (B) 9
(C) (C) 5
(D) (D) 17

5. If p and q are two prime numbers, then their LCM is:

(A) (A) 1

(B) (B) p

(C) (C) q

(D) (D) pq

6. Which of the following is a non-terminating repeating decimal?

(A) (A) $\frac{13}{8}$

(B) (B) $\frac{7}{16}$

(C) (C) $\frac{7}{15}$

(D) (D) $\frac{3}{10}$

7. The product of a non-zero rational and an irrational number is always:

(A) (A) Rational

(B) (B) Irrational

(C) (C) Integer

(D) (D) 1

8. If n is a natural number, then $9^n - 4^n$ is always divisible by:

(A) (A) 1

(B) (B) 2

(C) (C) 5

(D) (D) 13

9. The value of the exponent of 2 in the prime factorization of 144 is:

(A) (A) 2

(B) (B) 3

(C) (C) 5

(D) (D) 4

10. Given that $HCF(252, 594) = 18$, the $LCM(252, 594)$ is:

(A) (A) 8136

(B) (B) 8316

(C) (C) 8216

(D) (D) 8416